OT LAB EXAM / INTEGRATED OPTIMIZATION REPORT

Earthquake Inspection Optimization

Machine learning prediction · Quantum optimization · Parallel verification

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1 Problem formulation

Choose exactly three of eight locations. Let binary x_i (0 or 1) indicate inspection, $\hat{y}_i$ denote predicted damage and t_i the inspection time in hours.

Maximize $\Sigma_{i=1\ldots8}\ \hat{y}_i x_i$ **subject to** $\Sigma_{i=1\ldots8}\ x_i = 3$ and $\Sigma_{i=1\ldots8}\ t_i x_i \le 10$.

There are **C(8,3) = 56** unordered triples: **35 feasible** and 21 exceeding the time limit.

2 Working solution connecting all three domains

Machine learning. The exact seed-63 generator produces 120 records with density d, building age a, liquefaction index l and epicentral distance r. Ordinary least squares [1] matches its linear structure. A separate seed-7 split uses 96 training and 24 test records: **MAE 0.4137, RMSE 0.5260, R^2 0.9632**. Refit all records for the candidate predictions:

$\hat{y} = 5.129735 + 0.297040d + 0.080070a + 3.976314l - 0.125008r$.

Location	Predicted damage	Time (h)	Location	Predicted damage	Time (h)
L1	14.5218	4	L5	14.2236	4
L2	11.4994	2	L6	11.3326	3
L3	11.3384	3	L7	6.5657	3
L4	13.8944	5	L8	10.4114	3

Quantum computing. Encode the time inequality with four slack bits: $s = b_0 + 2b_1 + 4b_2 + 8b_3$. Minimize the 12-qubit QUBO: $E = -\Sigma\ \hat{y}_i x_i + A(\Sigma\ x_i - 3)^2 + B(\Sigma\ t_i x_i + s - 10)^2$.

$A = B = 1 + \Sigma|\hat{y}_i| =$ **94.7873**, exceeding the reward range for a unit violation. QAOA [2] starts from $|+\rangle^{12}$ and alternates cost-phase and Rx mixer gates. Depths 1, 2, 3, 4 and 6 use three starts each; the lowest-expectation run selects **p = 6**. A local simulator independently executes the exported H/Rz/CX/Rx circuit before **4,096 seeded measurement shots**; state fidelity agrees within 10^{-10}.

Parallel computing. Two spawned process workers [3] evaluate every triple. A sequential pass confirms the exact optimum; all 4,096 QUBO assignments and five verification test groups also pass.

3 Results and comparison

Method	Selected locations	Damage score	Hours	≤ 10 h?	Runtime (s)
Greedy top-3	L1, L4, L5	42.639800	13	No	0.000007
QAOA best feasible	**L1, L2, L5**	**40.244750**	**10**	**Yes**	**9.812743**
QAOA modal state	L1, L2, L8	36.432519	9	Yes	9.812743
Parallel exhaustive	L1, L2, L5	40.244750	10	Yes	0.075279

Wall times are single-run measurements. Quantum time includes encoding, all optimization attempts, simulation and sampling; parallel time includes process startup and reduction. ML preparation (0.005073 s) and report writing are excluded. Both quantum rows summarize the same run.

4 Interpretation and limitations

The **best feasible quantum sample matches exhaustive search**: **L1, L2, L5**, damage **40.244750**, time **10 h**, zero optimality gap. It appears **29/4,096 times (0.71%)**; feasible shots are 36.91%. The modal full state is L1, L2, L8; after aggregating slack outcomes, the modal location set is L7, L8 (infeasible). Best-feasible selection uses classical postprocessing; this is **not evidence of quantum advantage**.

Limitations: (1) Synthetic linear records do not establish real-earthquake accuracy; prediction uncertainty can change the choice. (2) Bounded QAOA has no convergence guarantee; statevector simulation scales exponentially and omits hardware noise.

References

[1] NumPy: numpy.linalg.lstsq. [2] Farhi, Goldstone & Gutmann (2014), A Quantum Approximate Optimization Algorithm.
[3] Python documentation: multiprocessing. Implementation: Sagarankita/Ankita-OT-Lab-Exam-, commit 8d9941c.